



openFinance API Framework

A Guide to Version 2.4

31 March 2026

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1 Introduction

With [PSD2] the European Union has published a directive on payment services in the internal market. Among others [PSD2] contains regulations on services to be operated by so-called Third Party Payment Service Providers (TPP) on behalf of a Payment Service User (PSU). These services are

Payment Initiation Service (PIS) to be operated by a Payment Initiation Service Provider (PISP) TPP as defined by article 66 of [PSD2],

Account Information Service (AIS) to be operated by an Account Information Service Provider (AISP) TPP as defined by article 67 of [PSD2], and

- Confirmation of Funds Service (COF) to be used by a Payment Instrument Issuing Service Provider (PIISP) TPP as defined by article 65 of [PSD2].

To implement these services (subject to PSU consent), a TPP needs to access the account of the PSU. The account is managed by another PSP called the Account Servicing Payment Service Provider (ASPSP). To support the TPP in accessing the accounts managed by an ASPSP, each ASPSP has to provide an "access to account interface" (XS2A interface). Such an interface has been defined in the Berlin Group NextGenPSD2 XS2A Framework.

The XS2A Framework is now planned to be extended to premium services. This interface is addressed in the following as **openFinance API**. This openFinance API differs from the XS2A interface in several dimensions:

- The extended services might not rely solely on PSD2 anymore.
- Other important regulatory frameworks which apply are e.g. GDPR.
- The openFinance API can address different types of **API Clients** as access clients, e.g. TPPs regulated by an NCA according to PSD2, or corporates not regulated by an NCA.
- The extended services might require contracts between the access client and the ASPSP.
- While the client identification at the openFinance API can still be based on eIDAS certificates, they do **not** necessarily need to be *PSD2-compliant eIDAS certificates*.
- The extended services might require e.g. the direct involvement of the access client's bank for KYC processes.

Note: The notions of API Client and ASPSP are used because of the technical standardisation perspective of the openFinance API. These terms are analogous to "asset broker" and "asset holder" respectively in the work of the ERPB on a SEPA API access scheme.



Note: In implementations, several ASPSPs might offer their services in an aggregation platform. Such platforms will be addressed in the openFinance API Framework as "API provider".

Note: At some places, the openFinance API Framework is still using the technical TPP notion also for the openFinance APIs. In this context, the TPP is not necessarily a licensed TPP as mentioned above, but for certain services may just be any "third party provider".

The following account access methods are covered by this framework:

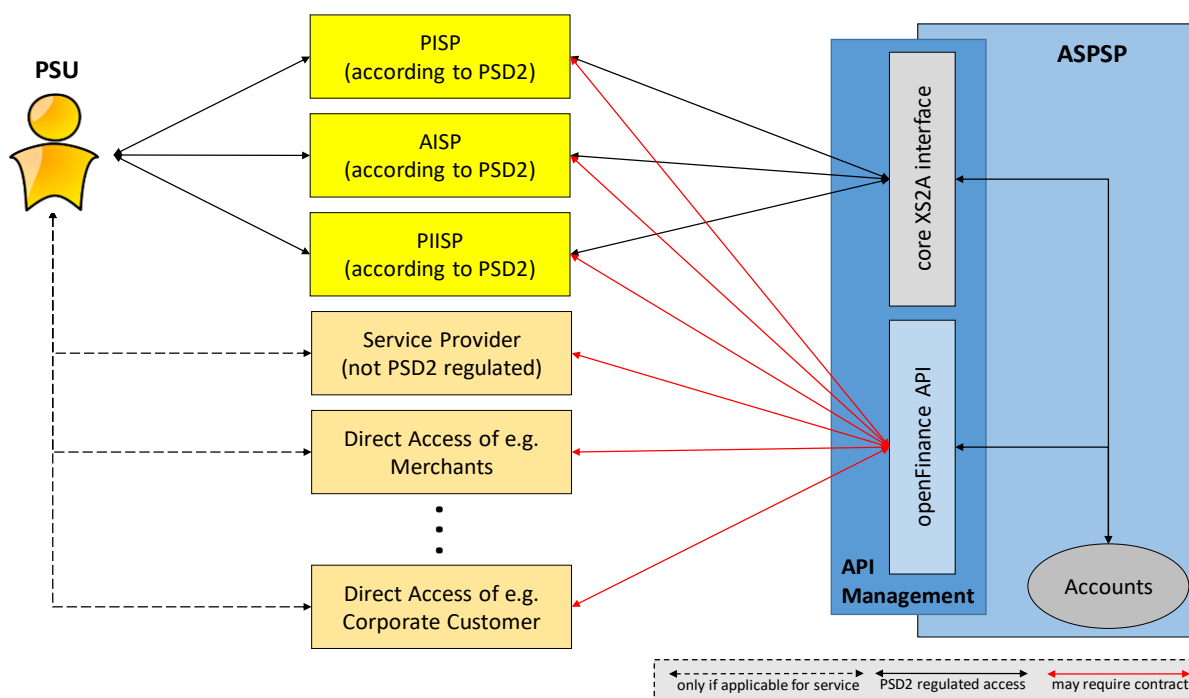


Figure 1: Core XS2A interface and openFinance API

The ASPSP may restrict access to the services offered at its openFinance API and requires dedicated onboarding. The requirements for the rights to access the services offered at the openFinance API are out of scope for this document. These requirements are in detail in [oFA-OR-Adm].

In contrast to the services of the openFinance API, the ASPSP has to offer access to the services of its core XS2A API to any TPP without discrimination as long as the TPP has the necessary licence to access a service as PISP, AISP or PIISP from an NCA according to the regulations of [PSD2].

The Berlin Group had started to publish first openFinance Service specifications in 2021 and 2022, which were technically based still on the NextGenPSD2 API V1.3.x. The Berlin Group reviewed this framework to adapt it better to the extended scope described in the new

framework. This review results in a Version 2 of the full openFinance API Framework. As part of the review, the documentation has been fully re-structured.

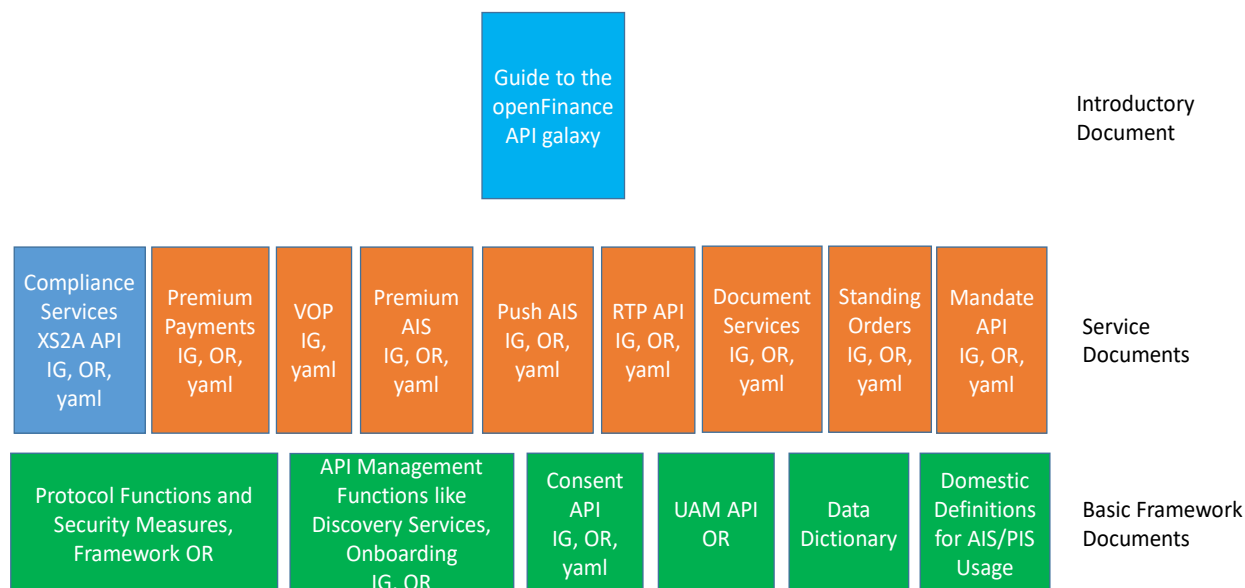
In between, the Berlin Group has extended existing services by more functions and published several new services. This document gives an overview of

- the new documentation,
- the new services covered,
- the major changes from NextGenPSD2 API Framework Version 1.3.x to openFinance API Framework Version 2.0, and
- the major changes from Version 2.0 to Version 2.1 as well as from Version 2.1 to Version 2.2, from Version 2.2 to Version 2.3 and from Version 2.3 to Version 2.4.



2 Documentation Overview

The openFinance API Framework V2.4 is structured in its documentation as follows:



For each service, the documentation is divided into 3 parts:

- Operational rules, which explain the business processes and business requirements for the underlying service.
- Implementation guidelines, defining the essential API messaging and flows.
- A yaml file with all messaging details.

2.1 Service Description

The service descriptions are divided into the descriptions for

- Compliance Services: [oFA-OR-Com] , [oFA-IG-Com]

These services are covering payment initiation (PIS), account information for payment accounts (AIS) and confirmation of funds on payment accounts (PIIS) based on the PSD2 requirements.

- Premium Payment Services: [oFA-OR-XPIS], [oFA-IG-XPIS]

These services are covering premium payment initiation services like payments with reservations of funds, payments with flexible execution and adapted amounts or recurring payments with adapted amounts or multiple recurring payments..

- Verification of Party (a.o. VOP) Services: [oFA-IG-VOP]

These services cover the verification of a party name or identification with the actual owner name or identification of an addressed party, which is confirmed by the account servicer of the addressed party. Besides a related API in the inter-PSP sphere, this document also addressed related VOP services offered as API from an ASPSP towards their own customers. In addition, the document specifies the integration of a VOP service into payment initiation services.

- Premium Account Information Services: [oFA-IG-XAIS]

These services are covering AIS for accounts different from payment accounts, i.e. single card accounts, loans and savings accounts as well as securities accounts. In the latest version, also the account statements for current accounts are added.

- Push Account Information Services: [oFA-OR-PAIS], [oFA-IG-PAIS]

In addition, a service to push AIS related information like account entries, balances or incoming request to pay to the API Client is covered. These services are also adding the possibility to provide not only transaction reports like in compliance PSD2 APIs, but also account statements.

- RTP Services: [oFA-OR-RTP], [oFA-IG-RTP]

The Request to Pay Services cover a pure messaging service to request a payment from a debtor via API, including an e-invoice. A consent or subscription to such a service might be mandated.

- Postal Services for Documents: [oFA-OR-Doc], [oFA-IG-Doc]

These services offer the possibility to post documents to the PSU mailboxes via the openFinance API. A consent or subscription to such a service might be mandated.

- Standing Orders Services: [oFA-OR-SO], [oFA-IG-SO]

These services offer to initiate periodic payments as standing orders with the additional benefit that all standing orders can be managed through one API Client even if the initial standing order was initiated through a different API Client or via Online Channels.

- Mandate API: [oFA-OR-Man], [oFA-IG-Man]

These services offer the possibility to authorise a direct debit mandate via the openFinance API.

2.2 Framework documentation

The underlying framework documentation is divided into the following parts:

- Protocol Functions and Security Measures: [oFA PFSM], [oFA-OR-FW]

The first document introduces notations and formats used throughout the openFinance API Framework. It covers authorisation processes (including signing baskets) as well as header parameters used in all services. It introduces message and body signing including encryption functions as well as a resource status notification function. In the upcoming version, it will also cover signed requests as SCA approach to prepare e.g. the usage of the openFinance API as corporate direct access. In addition, it will also contain the payment formats used in several service specifications.

The second document covers the description of all these framework functions on an Operational Rules level to support reading by e.g. product management staff without the related technical background.

- Consent API: [oFA-OR-Co], [oFA-IG-Co]

Explicit consent management via the API is needed not only for AIS but also for several premium services. These documents describe and specify the related consent API in detail.

- User and Account Management Services: [oFA-OR-UAM]

For corporate direct access, several data management functions need to be offered:

- Services to manage user certificates as well as user data, account access rights and payment authorisation limits.
- Services to manage existing accounts of a corporate as well as account details.

Corresponding detailed implementation guidelines will be developed by the end of 2026.

- API management functions: [oFA-OR-Adm], [oFA-IG-Adm], [oFA Disc]

For premium services, specific onboarding functions are mandated to ensure that e.g. scheme directory or fee information like price lists is properly handled on the API level. The related services will allow API Clients to post signed price lists to the API to automatise fee changes applicable to the API. In addition, detailed discovery services are standardised. These services shall ensure a compact documentation of the APIs provided by the ASPSP with technical, operational and business parameters.

- Data Dictionary:[oFA DaD]

A separate data dictionary describes all complex data types and code lists used within version 2.x of the openFinance API Framework. These data structures will be used in compliance as well as premium services. This document has been enriched quite far, by adding data models for the new services. In addition, the payment formats are enriched by the possibility to support hybrid address formats and some more query parameter have been added for AIS functions.

- Domestic formats: [oFA DomDef]

Payment data formats are separately documented with all adaption to SEPA payments as well as domestic payment instruments. Starting from Version 2.3, the payment formats as such will be integrated in [oFA PFSM] and only requirements on domestic AIS or PIS formats are stated in this document.



3 Major changes from Version 2.0 till Version 2.4

3.1 Major Changes openFinance API Framework from V2.3 to V2.4

Some major new framework functions are introduced in the latest framework documentation. The following changes have been applied to the underlying framework document [oFA PFSM] in version 2.4:

- A new category of service functions has been introduced which specifies the updating process of an existing transaction resource. This service initiates the change of an already authorised resource. The procedure of this service is detailed for a general case in the framework. In this case, a resource may obtain two different representations – one containing the currently authorised and active data and one containing the changed but not yet authorised data.

Both representations need to be addressable via API, since the active version is still valid and is only de-activated after the authorisation of the new and adjusted representation by the PSU (via SCA). As both representations must be addressable via the API, ETags as standard web methodology for dealing with new resource representations and organising the several representations of resources at the same time are introduced to the openFinance API Framework.

Moreover, to support the management of two representations of a resource via the API, several new header parameters and service functions have been added.

- Introduction of multi-channel resp. multi-API Client access to the openFinance APIs via the direct access for corporate customers and the API Client channels. Related service functions and operational rules to support sharing resources via multiple channels and/or multiple API Clients are added to the framework as well.

The changing processes of transaction resources in case of a multi-channel or multi-API Client access to the respective resources stress the necessary complexity of these new additions to the framework. Additional header parameters are introduced that should be used to simplify and ensure the functionality of the services in such scenarios.

Version 2.3 of the Framework has added the function of message body signing as an SCA approach, enabling the openFinance API Framework for corporate direct access. The integration into the technical API Framework description in [oFA PFSM] has been revised and restructured editorially for clarity reasons.

A new premium service allowing standing orders to be initiated and managed via the openFinance API has been included to API framework documentation, the related documents [oFA-IG-SO] and [oFA-OR-SO] are published for this service.

Several service specifications have been updated and now rely on the new Change Transaction Resource mechanism of the framework, among others this includes the new service function to prolong a multiple recurring payment, c.f. [oFA-IG-XPIS], and the inclusion



of a new subscription entry to an existing subscription for push account information services, c.f. [oFA-IG-PAIS].

3.2 Major Changes openFinance API Framework from V2.2 to V2.3

The new version 2.3 of the Framework documentation now includes a separate document [oFA-OR-FW] containing the description of the technical functions as defined in [oFA PFSM] on a product properties level to support readability of the overall openFinance API Framework documentation.

Additional documents [oFA-OR-Co] and [oFA-OR-Com] are published to provide a better understanding on product properties level, the business processes and business requirements of the premium services of the consent API and the compliance API.

The payment format definitions used for [oFA-IG-XAIS], [oFA-IG-XPIS] as well as [oFA-IG-RTP] are removed from [oFA DomDef] and integrated into [oFA PFSM] to enable a clear release control for payment formats as well.

The openFinance API Framework in now also support a.o. digital signatures as an SCA approach. For now, this SCA approach is only included for the direct access of corporate customers of ASPSPs via the openFinance API.

3.3 Major Changes openFinance API Framework from V2.1 to V2.2 resp. V2.3

Several new services have been added on business service as well as framework service level, see above. For several services, which have been announced in summer 2024, the documentation has been finalised and published in-between.

The following changes have been applied to the underlying framework document [oFA PFSM] in version 2.2:

- The structure of the framework documentation has been optimised by introducing two different natures of services, one of which is related to transaction initiations (existed already) and the other is related to the creation of service resources.

In more detail, the description of a service resource, the header parameters of a service resource creation request and the links of the response of the service resource creation are correspondingly added.

- The requirement not to restrict redirect URIs to TPP domains protected by the QWAC has been detailed for PSD2 compliance services.



- Some new features have been added to support the integration of a Verification of payee function into payment initiation APIs.
- The examples for signing messages have been completely reworked.

In addition, a new version 2.3 of the Framework documentation is announced for autumn 2025 which will address the following points:

A description of the technical functions as defined in [oFA PFSM] on a product properties level will be added in [oFA-OR-FW] to support readability of the overall openFinance API Framework documentation.

In addition, the payment format definitions used for [oFA-IG-XAIS], [oFA-IG-XPIS] as well as [oFA-IG-RTP] will be integrated into [oFA PFSM] to enable a clear release control for payment formats as well.

The openFinance API Framework in its next version will also support a.o. digital signatures as an SCA approach, in the beginning for the support of direct access of corporate customers of ASPSPs via the openFinance API.

3.4 Major Changes openFinance API Framework from V2.0 to 2.1

First, the framework documentation has been completed and more business services have been integrated as listed above. All technical functions like status notification and signing baskets have been adapted to cover also the new services.

In addition, the following points have been adapted besides many editorial clarifications, clarifications on code usage and many adaptations on attribute level.

- All header parameters with "TPP-" prefix have been adapted to "Client-" prefix. The reason behind that is further preparatory work for the usage of all APIs also in a corporate direct access mode, i.e. directly between banks and their corporate customers. A standardisation of such APIs on European level is seen as very important e.g. for all real-time use cases. The re-usage of the PSD2 infrastructure for the related direct access scenario will offer lots of implementation and documentation synergies.
- The framework got a new option to select a debtor IBAN in payment APIs in the decoupled and embedded SCA approach. This is a more efficient solution for the "No manual IBAN entry" requirement from the regulators and the market.
- The message signing algorithm using QSeals certificates has been adapted to also protect URI, i.e. to protect e.g. payment resources. Such a change was needed, since the new services come with payment initiation calls with a prior PSU authorisation, which comes with new technical risks to be mitigated.
- The framework will also offer to manage the QSealCs only via the dedicated onboarding endpoints, where API Clients will be mandated to use key references

regarding the QSealCs applied within the business services. This will enable a more efficient certificate evaluation within the ASPSP.

- The framework offers a new SCA approach "ASPSP-Channel". This is an approach, where the authorisation follows later in the ASPSP online channel. This SCA approach is only applicable to premium services. The API Client then still might change its view and authorise the transaction later on via the API, e.g. if the PSU is in a session with the API client in-between. Please note that this asynchronous authorisation is different from the RTP authorisation since the SCA channel can be dynamically changed and the resource is fully addressable and under control in the new SCA approach (differently to RTP, where the control is solely on ASPSP side and the related payment resource is not retrievable in the API).



4 Major Changes in the transition from NextGenPSD2 1.3.x to openFinance 2.x

The following major changes are introduced in the framework, when migrating from Version 1.3.x to Version 2. Please note that the openFinance API Framework cannot mandate any implementation transition plans from Version 1.3.x to Version 2.0.

- Compliance Services

Header parameters with prefix "TPP-" have been changed to the prefix "Client-" due to the fact that the openFinance APIs are intended to be used also for direct access by corporate customers in a next step. Only header parameters which are needed in a third party model only like "TPP-Redirect-URI" keep the "TPP-" prefix.

The attribute "tppMessages" has been renamed to "apiClientMessages" to recognize this also on message body level.

A new header "TPP-SCA-Approach-Preference" has been added to combine the semantics of the two headers "TPP-Redirect-Preferred" and "TPP-Decoupled-Preferred" in one entry.

psuName is not any more delivered in transactionStatus in PIS nor on account details level in AIS but only within the status of the authorisation sub resource. The reason is that this element is a PSU related information and not payment or account information.

The function to submit a second password has been removed from the services.

The AIS / PIS session combination has been removed. It is now covered by introducing the access right "paymentInitiations" to the consent for optional support by the ASPSP.

ISO20022 Reason codes specifically for the transactionStatus "Rejected" have been introduced. This is replacing the former proprietary message codes in some situations.

If a cancellation of a payment needs an authorisation, the related authorisation process can be an implicit authorisation as for payment authorisation itself. The same applies to authorising a signing basket.

- Consent API

The access attribute has been structured in more detail. It is now separating different account categories explicitly and is relating access rights to each single account.

The consent type is now an explicit attribute "consentType" in the request, better separating the detailed, aspspManaged, accountList and global consent type. A dedicated message code "CONSENT_TYPE_NOT_SUPPORTED" has been introduced for the case of a mismatch.

The attribute "validUntil" was re-named into "validTo".

A new consentStatus has been introduced: "replacedByTpp" indicating explicitly that a consent resource has been invalidated by a TPP by submitting a new consent.

The ASPSP is required to offer in the consent resource also for the aspspManged and global consent all account references and links to the related APIs after successful SCA. Thus reading the consent resource after PSU authentication will provide a lot of information to the API Client. This was introduced to save pings in the API when several account categories are covered.

The extended service defined for confirmation of funds was adapted to the new consent data model for harmonisation reasons.

The optional extended Multiple Consents Service was integrated into this document.

A new premium consent category "user parameter access" is supported. The related discovery function for user parameter like signing rights, limits etc. will be introduced later.

- Basic framework

Protocol functions like the signing basket function or the resource status notification has been added to this document. The authorisation descriptions are now provided generically for all business transactions (yielding to related API resources after submission) supported in the API.

In the whole document, the situation of API Clients being a TPP in a compliance PSD2 scenario or any access client to an ASPSP API is differentiated, e.g. in certificate usage.

Message signing has been migrated from the sign http procedure to the ETSI JADES profile described in the OBE signing proposal.

A rule for interval borders related to attributes with "From" or "To" as suffix has been added to clarify that the intervals are including the border values.

- Data dictionary

The payment formats and account reports for Version 2.0 have been adapted to the formats already published in 2022. The reason was to integrate all requirements collected during the last years from several countries and several premium services in one unified data structure.

The full structured ISO20022 address format is supported now.

- Payment formats

See above.

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